



ICT and productivity: conclusions from the empirical literature



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ABSTRACT

Innovation drives economic competitiveness and sustained long-term economic growth. Especially the emergence and intensive utilization of information and communication technologies (ICTs), which spawned the beginning of the digital economy two decades ago, heavily affected the opportunities and efficiency of how firms produce and provide goods and services. We provide an overview of the empirical literature on ICT and productivity and highlight the main results and methodological differences. The majority of studies indicates that the productivity effect of ICT is indeed positive and significant. However, methodological approaches of how to appropriately estimate the ICT effect matter. While aggregate and sectoral growth accounting exercises suggest stronger differences of the ICT effect between US and Europe, firm-level analyses suggest no significant country differences. Moreover, we shed light on the notion of ICT being a General Purpose Technology (GPT) enabling further innovations. Most of the GPT evidence on ICT is found for the US, while evidence for European countries is harder to come by. However, more theoretical and empirical research is needed to better understand spillovers and externalities of ICT and how these technologies transform our economies.

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1. Introduction

The wealth of nations and their economic fortunes are ultimately driven by productivity. This is because more productive workers earn higher wages and enjoy higher living standards. Hence, determining the ultimate driving forces of productivity is an important question for researchers and policymakers alike. One fundamental source of productivity is innovation. Not all least since [Solow \(1957\)](#) who highlighted that sustained long-run growth of per capita GDP depends on technological progress, the New Growth Theory literature ([Romer, 1990](#); [Aghion and Howitt, 1992](#); [Grossman and Helpman, 1991](#)) emphasized the forces that generate technological change. The key insight of this literature is that it requires ever

more efficient use of available resources to produce sustained long-term economic growth through processes of technological development and innovation. Innovation in particular is itself fueled by research and development. As long as there is enough inventive potential in an economy and no severe distortions related to the creation of new products, the economy will grow.

Determining the driving forces of productivity becomes even more important since Europe's productivity grew slower than in the US during the mid-1990s lasting until the mid-2000s. A broad consensus attributes a large share of this to strong investments in information and communication technologies (ICTs), and much of it originated in industries that produced and used ICT intensively. ICT is considered a special case of new technologies that serve as enabling technologies leading to even further innovations. However, not all developed countries shared the same benefits from ICT as the US. In case of Europe, capital intensity of ICT started to lag behind the US during the same time US productivity growth took off.

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The necessity to foster productivity growth gave rise to the adoption of corresponding ICT policies. In the US the stimulus package “American Recovery and Reinvestment Act” from 2009 set \$7.2 bn in grants to invest in broadband and wireless internet access and another \$290 m to upgrade IT platforms at the State Department¹ with the goal to reshape “our economy to compete in the digital age” as the then-presidential candidate Barack Obama declared in his announcement speech in 2007.² The European Commission also declared that “Europe needs download rates of 30 Mbps for all of its citizens and at least 50% of European households subscribing to internet connections above 100 Mbps by 2020. The Digital Agenda aims to turn this ambition into reality by stimulating investments and proposing a comprehensive radio spectrum plan”.³ These commitments show that the importance of ICT for economic growth has convinced policymakers on both sides of the Atlantic. Nevertheless, sceptics remain about the exact extent of ICT’s impact and its role in reshaping the economy. Carr (2003) argued that Information Technology (IT) due to its ubiquity has become a commodity and advises to spend less and follow rather than lead with IT investments. Gordon (2010) also questions the extent of the ongoing economic impact of ICT stating that the problem “of the ICT revolution is that it is increasingly burdened by diminishing returns”. Our survey will therefore gather the academic empirical evidence on this question and summarize the key conclusions about the effect of ICT on productivity.

Besides its productivity enhancing effects, ICT has become a substantial part of the social and business environment. Massive reduction in computing and communications costs (Jorgenson, 2001) has spurred huge investments (Colecchia and Schreyer, 2002) and engendered a substantial restructuring of the economy (Brynjolfsson and Hitt, 2000). Computers, mobile communication devices and the internet are integrated into everyday lives and are consequently changing how business is done and markets work in manifold ways. It is therefore not surprising that ICT attracts academics studying how exactly ICT is changing the economy. While this leads to insights into how ICT affects production processes, efficiency and output growth, the abundance causes confusion specifically arising from a broad literature on different aggregation levels, ICT products and methods, as well as seemingly contradictory findings and various numbers quantifying the ICT effect. In fact, the number of ICT studies has already triggered several reviews as summarized in Table 1.

These review papers give us a big picture of the field. Firstly, in the early days explanations for and discussion of

the Solow Paradox⁴ dominated and shifted to the question of causes for the US productivity resurgence in the late 1990s later on. Secondly, measurement issues are of big importance to ICT studies, mainly dealing with the correct measure of ICT inputs but also the right output measure. Thirdly, organizational complements and intangible assets are considered crucial to ICT’s influence on productivity, which are even harder to pin down with empirically. And fourthly, there is a big separate field of studies investigating communication technologies and their specific effect which is not dealt with specifically within ICT review studies. While some ICT studies incorporate communication technologies in their ICT measure, this is rarely dealt with explicitly.

These surveys offer orientation for the large literature field but the growing number of studies and often seemingly contradictory results make it hard to keep track of the state of evidence. We therefore want to provide a guide and give a structured picture of the existing literature. We focus on two specific questions which distinguish our review from existing reviews. Firstly, what is the size and existing range of quantitative results in the empirical literature of the ICT effect on productivity? To answer this, we restrict ourselves to comparing consistent estimates and therefore only include studies with output measures on the left-hand side. Alternative measures of output like consumer surplus will be discussed in more detail in the last sub-section of chapter 3. Secondly, we will discuss the state of the evidence on the “General Purpose Technology (GPT) hypothesis”, i.e. whether ICT is a normal asset or can be classified as a GPT. The latter would imply that ICT acts as an enabling technology for further innovations, by which it exhibits an influence on economic growth and productivity beyond the effect of regular capital goods. This is especially relevant for policy decisions as it lays the foundation for public investments. Measures to rectify underinvestment are only justified in the presence of spillovers.

To quantify the impact of ICT on productivity growth and evaluate the GPT hypothesis, we classify the existing literature by three different categories:

- **Method:** we distinguish non-parametric approaches, primarily growth accounting exercises, and parametric techniques, like econometric estimations of production functions.
- **Aggregation level:** country, industry and firm level.
- **ICT product/measure:** IT (hardware/software), data communication (internet/broadband), telecommunication (mobile), and ICT (in its most extensive definition), specifically including studies related to communication technologies.

Since we focus on studies with output or productivity as the dependent variable and its numerical results, we do not include studies on consumer surplus (e.g. Greenstein and McDevitt (2009); Brynjolfsson and Hitt (1996b)), which are not about value creation, but rather on its distribution. We also exclude studies that discuss firm performance indicators such as return on asset, market share or market value (e.g. Gattiker and Goodhue (2005); Tam (1998)). Further, studies that measure the effect of ICT on employment, wages or innovation are not part of our survey.

¹ For detailed information on the American Recovery and Reinvestment Act of 2009 see: <http://www.recovery.gov>.

² For full speech go to <http://obamaspeeches.com> and “Announcement For President, February 2007”.

³ European Commission’s official statement on the Digital Agenda 2010–2020. http://ec.europa.eu/information_society/newsroom/cf/pillar.cfm?pillar_id=46.

⁴ The Solow Paradox goes back to an article written in the New York Times by Robert Solow, which culminates in the much repeated quotation “You can see the computer age everywhere but in the productivity statistics”, the scanned original article can be viewed at www.standupeconomist.com/blog/economics/solows-computer-age-quote-a-definitive-citation.

Table 1

Overview of previous literature reviews on ICT contribution.

Study	Method	Results
Brynjolfsson and Yang (1996)	Written survey based on over 150 studies	Discusses explanations for the productivity paradox, measuring the IT output link was practically impossible due to lack of data and use of inadequate analytical methods
Brynjolfsson and Hitt (2000)	Literature survey on how IT is linked to higher productivity and organizational transformation, based mainly on firm-level studies	IT performance depends on complementary organizational investments and these investments lead to improvements in intangible aspects. These factors are not well captured by traditional macroeconomic measurement approaches, hence generating the Solow Paradox
Baily (2002)	Summarizes growth accounting and case study evidence and assesses other indicators of structural change	IT is an important, but not the only cause of the productivity resurgence in the 1990s. Competition and globalization were further basic drivers
Dedrick et al. (2003)	Written survey on 19 firm-level and 15 country-level studies between 1987–2002	Productivity paradox refuted, wide range of IT investments among different organizations can be explained by complementary investments in organizational capital
Melville et al. (2004)	Develop a model of IT business value based on resource based view to review the literature	IT investments provide value, but impact depends on levels of complementary resources, competitive climate, general macroeconomic environment. Synergies between technical and human IT resources yield short lived competitive advantage
Stiroh (2005)	Meta-Analysis (20 studies from 1994–2002)	Study characteristics explain about 35% of the variation in the IT elasticities. Median elasticity at 0.046
Draca et al. (2008)	Survey micro and macro literature	Macro studies meanwhile show evidence of ICT impact. In micro studies the effect is larger than the neoclassical contribution would expect, which is due to organizational complements
Holt and Jamison (2009)	Literature survey on broadband studies	Broadband has positive impact, but cannot be measured with any precision

Our survey records results and methods of about 150 studies, which break up into 73 studies at the macro or country level, 38 at the industry level and 47 at the micro or firm level. For details of the literature search procedure see Section 5.2. Our main findings show that there is a substantial variation in ICT elasticities depending on the methodology. While country- and industry-level growth accounting exercises show diverging ICT effects for the US and Europe and a stronger impact of ICT on productivity growth in the US, econometric firm-level studies show no significant country differences. Moreover, the latter provide solid evidence that over the last two decades an increase of ICT investment by 10% translated into higher output growth of 0.5–0.6%. The empirical evidence on the GPT hypothesis of ICT is mixed and needs further investigation.

2. Is ICT a GPT?

One aspect of the ICT story is to observe innovations in the IT-producing sector, falling prices for IT products and the resulting investments in ICT (Jorgenson, 2005). Another aspect is whether apart from these phenomena ICT investments go beyond conventional capital equipment leading to concepts associated with being an “enabling technology” (Jovanovic and Rousseau, 2005). This may be especially convincing given that knowledge has become increasingly important to economic activity (Powell and Snellman, 2004) and that the main function of ICT is storing and sending information, i.e. knowledge. Further, through facilitating communication ICT promotes the creation of new knowledge through more efficient processes of collaboration (Forman and van Zeebroeck, 2010).

The enabling property of ICT is often illustrated with ICT investments in firms. These investments allow faster information processing and firms to think of new ways of communicating with suppliers or customers or arranging new distribution systems. Internal processes can be streamlined, reducing capital needs through better utilization of equipment and reduction in inventories and thereby space requirements. Increased communication and more timely and widespread transfer of information reduces coordination costs, the number of supervisors required reducing labor costs and facilitate better decision making (Arvanitis and Loukis, 2009; Atrostic et al., 2002; Gilchrist et al., 2001). Lower communication and replication costs will help businesses innovate through new products (Brynjolfsson and Saunders, 2010). But not only firms, also individuals can seek better prices for consumption and improve job search via the new communication technology, increasing overall efficiency throughout the economy. Communication technologies lower the fixed costs of acquiring information and the variable costs of participating in markets (Norton, 1992; Leff, 1984).

In the early days of the New Economy, another aggregate ICT effect supports the GPT character of ICT. While strong ICT investments in ICT-producing sectors spurred US productivity growth post-1995, there is some indication that efficiency gains from implementation of more productive ICT equipment was not limited to this sector only. Especially during the second, broader surge of the US economy post-2000 Stiroh and Botsch (2007) find that efficiency gains spilled over from ICT-producing sectors to industries that heavily used ICT. It is particularly this latter characteristic that makes ICT a GPT, since computers and related ICT equipment are expected to be utilized in most sectors as ICT diffuses.

All of this suggests that ICT is a General Purpose Technology (GPT) (David, 1990; Bresnahan and Trajtenberg, 1995; David and Wright, 1999; Jovanovic and Rousseau, 2005). The characteristics of a GPT have been outlined as:

- Applicability across a broad range of uses, “Pervasiveness”.
- Wide scope for improvement, experimentation and elaboration, keep lowering cost within the producing and using firms “Improvement”.
- Makes it easier to invent and produce new products and processes within the producing and using firms “Innovation spawning”.

These considerations are closely related to spillovers, where social returns exceed private ones Jaffe et al. (1993). If ICT is a source of spillovers, the second and third conditions are not only given in ICT-producing sectors, but spillovers exist from producers to users of ICT and may even feed back to producers. These kind of spillovers were termed “vertical” spillovers by Bresnahan and Trajtenberg (1995). Further, if other firms can learn from the experimentation and elaboration efforts of others, we can observe “horizontal” spillovers. For example, newly developed management ideas or organizational knowledge related to the adoption of ICT could diffuse among firms. In the most basic case spillovers can be direct network externalities, when the simple adoption of further users enhances the benefit of having also adopted the technology. Jointly, the three GPT characteristics point to a time consuming process, so that it can take decades to fully materialize and transform business processes accordingly to reap the full benefits (David and Wright, 1999).

The main difference between the conventional view on the productivity effects stemming from ICT and those postulated by the GPT hypothesis culminates in the two following views. While the former assumes that ICT increased productivity mainly through capital deepening and the input substitution of more for less productive inputs triggered by the fall in ICT prices and increased quality of semi-conductors and computers, the latter assumes that ICT has computerized businesses and the economy as a whole, leading to ever more innovation and increased productivity in ICT-using and -producing sectors. According to the conventional view, no TFP accelerations take place outside the ICT production sector. ICT productivity effects occurred through substitution of inputs of different marginal products within the using firms, while spillover effects and shifts of production functions of the using sectors through ICT are not considered. The empirical

evidence for and against the GPT case is assessed in Section 5.3.

3. Comparing the empirical approaches

We focus on studies with a direct output measure, e.g. sales at the firm level, GDP at the country level, or productivity. As productivity describes the relationship between output and the inputs that are required to generate that output, this allows for different measures for inputs and outputs. The most common productivity measure is labor productivity, which relates output to labor input and can be based on gross output or value added as the relevant output measure. Labor productivity usually is calculated by employing hours worked or number of employees for labor input. Using hours worked generally gives a more accurate measure of the actual economic activity. Other input measures use quality-adjusted hours worked (labor services) and thus additionally account for shifts within the skill composition of the labor force, incorporating the productivity effect of human capital. For capital, capital stocks or quality-adjusted capital stocks (capital services) are used, where the latter explicitly account for productivity differentials in asset types. Both labor and capital are single factor productivity measures, while simultaneous use of capital and labor inputs gives multi-factor or total factor productivity measures. For an overview of potential productivity measures see Table 2.

As the output measure either is gross output or value added, in the latter case intermediate inputs – energy, materials and purchased services – are subtracted from gross output. Value added is considered the more accurate measure of the actual surplus created. If gross output is used, intermediate inputs are included as an additive summand to avoid double counting over the value chain.

The starting point for most empirical productivity calculations or estimations is the Cobb–Douglas function, which postulates the relationship in its most common version between input and output as:

$$Y = AK^{\alpha}L^{\beta} \quad (1)$$

where Y is output, K capital, and L labor input, and capital and labor are weighted by their factor elasticities α and β , accordingly. A is total factor productivity (TFP), which will be discussed in detail in Section 3.1. This functional form implies Hicks-neutral technical change, for which the ratio of marginal products remains constant for a given capital-labor ratio (Barro, 1999).

Table 2

Overview of the main output and productivity measures. Source: Schreyer and Pilat (2001).

Output measure	Input measure labor	Capital	Capital and labor
Gross output	Labor productivity (based on gross output)	Capital productivity (based on gross output)	Capital-labor MFP (based on gross output)
Value added	Labor productivity (based on value added) Single factor productivity measures	Capital productivity (based on value added)	Capital-labor MFP (based on value added) Multi-factor productivity (MFP) measures

3.1. Growth accounting

The different approaches to assess the impact of ICT on productivity growth are either parametric or non-parametric. The most common non-parametric technique is growth accounting based on the paper by Robert Solow on technical change and the aggregate production function (Solow, 1957) and discussed in Barro (1999) and Aghion and Howitt (2007). Non-parametric approaches employ properties of the production theory to determine empirical measures of the parameters of a production function by constructing economically defined index numbers. A prerequisite for an exact identification of the parameters is the validity of neoclassical assumptions, like the possibility to represent production processes by production functions, separability of inputs, competitive factor markets (price takers) and efficient producer behavior (profit/cost maximization/minimization). For parametric techniques, econometric methods are applied to estimate the parameters of a production function directly. Parametric approaches will be discussed in detail in Section 3.2.

Growth accounting provides a well-established tool to examine how much of a country's or industry's growth in output can be explained by growth in inputs, especially by growth in different types of capital input. Steady advances in productivity measurement of input factors over the last decades nowadays enable researchers to distinguish between appropriately measured ICT and non-ICT capital based on national statistics (OECD, 2001). Researchers established growth accounting as a well tested approach to quantify the ICT contribution to output and productivity growth (Jorgenson et al., 2005; Jorgenson and Timmer, 2011; van Ark et al., 2003; Inklaar et al., 2005; Timmer and van Ark, 2005; Inklaar et al., 2007). The strength and intuitive appeal of the methodology lies in determining the sources of output and productivity growth according to a well designed and consistent economic framework. Nevertheless, it does not allow for establishing causal relationships. Also, the strong assumptions underlying the theoretical framework affect the interpretation of productivity and technology measures.

To construct an index that measures the contribution of combined inputs, the growth rates of the inputs factors (i.e. capital and labor) have to be weighted appropriately. Following production theory and simplifying assumptions, factor income shares are typically used as weights. These income shares approximate production elasticities, i.e. in case of labor, for example, the shares resemble the share of labor compensation in total costs. Similar to output elasticities in econometric estimations of production functions, these elasticities are interpreted as the effect of a 1% change in a specific input on output growth. Starting from a Cobb–Douglas production function with ICT/non-ICT capital and labor as inputs and TFP (A), the function can be log-linearized and differenced and ultimately divided by hours worked on both sides to generate a growth accounting equation in exponential growth rates:

$$\Delta \ln y = \bar{v}_{K_{NIT}} \Delta \ln k_{NIT} + \bar{v}_{K_{IT}} \Delta \ln k_{IT} + \bar{v}_L \Delta \ln L_Q + \bar{w}_{NIT} \ln A_{NIT} + \bar{w}_{IT} \ln A_{IT} \quad (2)$$

where $\Delta \ln y$ resembles labor productivity growth (output per hour worked), $\Delta \ln k_{NIT}$ non-ICT capital deepening (non-ICT capital input per hour worked), $\Delta \ln k_{IT}$ ICT capital deepening (ICT capital input per hour worked), and $\Delta \ln L_Q$ labor quality growth (growth in labor composition). The contributions to labor productivity growth are derived by weighting each of the factor growth rates by their respective weights $\bar{v}$ (two-period averages), which correspond to the revenue share of each factor in total output. Given the assumptions of perfect competition and constant returns to scale these input shares of capital provide an exact measure of the elasticity. In case of aggregate capital the elasticity can be decomposed into the following parts $v_K = \frac{rK}{Y}$, with K being the amount of capital, Y national income, and r the user cost of capital or marginal product of the capital. Consequently, rK is the share of capital income. The user cost of capital reflects how much would be charged for one period rental of a capital good. Due to rapid depreciation rates and strong price deflation, computers' rental prices are much higher compared to other assets (Colecchia and Schreyer, 2002). The elasticity of labor can be derived analogously as $v_L = \frac{wL}{Y}$, with L being the amount of labor employed, Y national income, and w the marginal product of labor, usually represented by labor compensation. Hence, wL represents aggregate labor income. This weighting procedure for both capital and labor implicitly assumes that inputs are paid the value of their marginal productivity.

The term A in Eq. (2) is TFP or the Solow residual, and exhibits positive growth rates whenever the growth rate of the volume of output rises faster than the growth rate of all combined inputs. Rather than interpreting the residual as pure technological change, TFP more precisely resembles a host of unobservable factors that affect the improvement in overall efficiency of how output is produced (Schreyer, 2001). Thereby, TFP does not capture the induced effects of technological change on growth, which are already captured by quality improvements of capital and labor and which is referred to as embodied technological change (Jorgenson, 1966). In fact, TFP represents all those technological improvements not directly captured by quality changes and thus represents disembodied technological change. Aghion and Howitt (2007) argue on innovation embodied in capital that “it is hard to separate the influence of capital accumulation from the influence of innovation”. In fact, because the quality-adjusted real price decrease is difficult to assess precisely and national income accountants tend to be careful regarding any data manipulation, the usual adjustments are too cautious resulting in too low TFP in upstream markets and vice versa for downstream industries.

In Eq. (2) residual growth is separated into TFP growth from non-ICT producing sectors ($\Delta \ln A_{NIT}$) and ICT-producing sectors ($\Delta \ln A_{IT}$). Therefore, industry TFP growth rates are weighted with their share of industry output of non-ICT producing and ICT-producing sectors in total output, $\bar{w}_{NIT}$ and $\bar{w}_{IT}$, respectively.

The impact of ICT in the neoclassical growth accounting framework is theoretically underpinned by the idea that an increase in ICT capital deepening (i.e. more ICT investment per employed labor input) is triggered by rapidly

falling IT prices. This was especially the case during the rise of the New Economy. In addition, rental prices of ICT are affected by rapid depreciation, which – according to Jorgenson's user cost of capital – raises rental costs of ICT relative to other assets. Hence, ICT capital must have larger marginal products to cover the high rental prices than other capital assets.

However, the main critique of growth accounting is in not explicitly accounting for the underlying causes of growth (Baily, 2002). Rather, it allows for the quantification of the proximate sources of growth in a systematic and consistent established framework. But in its most original form the framework does not consider a number of specific factors influencing growth including adjustment costs, inefficiencies, variable factor utilization, deviations from perfect competition and constant returns to scale, outsourcing and offshoring, management expertise, or intangibles that are omitted from official data accounts (Oliner et al., 2007). As TFP is measured as the residual in the equation, it is especially prone to mismeasurement. This error can be severe, so that Bresnahan (1986) concluded that TFP may just be an accounting identity rather than a measure for technical change. Because of the variable factor utilization and non-constant returns to scale TFP measurement has a procyclical bias (Basu and Fernald, 2001). Another important shortcoming is the lack of accounting for intangible capital, like organizational skills, production skills or specific worker training. These costs are treated as a current expense and therefore do not show up as a stock in balance sheets. These problems are widely discussed and recent studies have included a measure for intangible capital (Marrano et al., 2009) and additionally adjusted for utilization and adjustment costs (Oliner et al., 2007). However, more recently growth accountants have tried to overcome these conceptual hurdles. The growth accounting results of our survey are discussed in Section 5.1.

3.2. Productivity estimations

Using parametric approaches to estimate production functions avoids imposing a theoretically based relationship between production elasticities and income shares. In doing so the output elasticities of input factors are directly estimated from the data generating process, without assuming profit/cost maximization/minimization behavior of producers and competitive factor markets. Further, econometric estimations of elasticities enable deviation from the concept of constant returns to scale, an assumption that is only realistic at the firm level. Also, further assumptions of the growth accounting approach, like variations in factor utilization and adjustment costs (Fernald et al., 2001; Oliner et al., 2007) can be relaxed. In the first case changes in utilization of factors affect measured productivity as well as measured contributions of inputs. For ICT, this means that there could be a correlation with increased productivity without any “real” effects ICT. Usually productivity researchers try to correct for this by explicitly decomposing productivity measures into trend and cyclical components or by applying averaged periods over the business cycle. Particularly, the application of econometric models – given long enough time periods – render

accounting for business cycle effects manageable. Not accounting for adjustment costs in growth accounting exercises may lead to biased productivity effects from inputs because as new technologies are implemented and adjustment costs increase, productive resources are diverted from production leading to a temporary reduction in output.

Like the growth accounting equation, the regression equation is derived from log-linearizing the Cobb–Douglas production function and is estimated as follows (Brynjolfsson and Hitt, 1995, 2003):

$$\ln Q_{it} = \beta_1 \ln C_{it} + \beta_2 \ln K_{it} + \beta_3 \ln L_{it} + \text{controls} + \epsilon_{it} \quad (3)$$

where Q represents firm or industry value added, while for country studies GDP is typically used. C is ICT and K non-ICT capital, $\ln L$ is labor input. The overwhelming majority of the production functions are estimated with panel data. The index i stands for the observational unit, which according to the employed aggregation level either is a country, an industry or a firm, t represents the time period, most commonly measured in years. The usual control variables are time dummies or industry, region and age controls in firm-level studies. Some studies control for the observational unit in fixed-effects models to capture any time invariant idiosyncratic productivity effect. This may be particularly important as some firms or industries are *per se* more productive because of unobserved heterogeneity, e.g. because of better management practices and/or market structures. In some studies ICT capital has been approximated via penetration rates, e.g. in case of telecommunication capital the number of telephone per inhabitant or in case of IT capital computers per workers were employed.

The coefficient β_1 resembles the output elasticity of ICT and corresponds to the $\bar{v}_{K_{IT}}$ calculated from the national accounts in growth accounting exercises (see Eq. (2)). One benefit of estimating the elasticity econometrically is formally testing its statistical significance instead of postulating it by economic theory. Due to endogeneity issues this test has limitations however. One of the main critiques of ICT estimations is that it determines a correlation rather than a causal effect on productivity. This is because investment in ICT may be considered a driver, but also a result of productivity and economic growth. This endogeneity (or reverse causality) problem cannot be addressed in a simple regression model. Hence, another advantage of econometrically estimating production function elasticities is that depending on the econometric approach the issue of endogeneity may be explicitly addressed. Firm-level studies, for example, exploit the panel structure of the data by using dynamic panel data models that utilize lagged values of the ICT variable as valid instruments (Bloom et al., 2010; Brynjolfsson and Hitt, 1995; Hempell, 2005b; Tambe and Hitt, 2011). In doing so, the lagged variables of period $t - 1$, $t - 2$ and so forth are expected to correctly identify the causal effect of ICT on productivity in period t since all shocks occurring in t are no longer correlated to lagged ICT. Another possibility is to exploit time-series behavior of other inputs to make corrections for estimates of capital coefficients with the Levinsohn–Petrin estimator (Levinsohn and Petrin, 2003; Tambe and Hitt, 2011). Alternatively, production functions can be estimated in a

structural equation model that takes into account different channels among the variables and thus explicitly models endogeneity (Koutroumpis, 2009; Röller and Waverman, 2001). An alternative approach is adopted in Czernich et al. (2011), who estimate the ICT variable (broadband in their case) with a first-stage diffusion equation.

The Cobb–Douglas functional form is very restrictive in assuming a substitution elasticity among its inputs equal to one and homotheticity, which indicates that the relative demands for inputs are independent of the level of output. Therefore other more flexible functions have been proposed for estimation (Berndt and Triplett, 1991; Mefford, 1986). The constant elasticity of substitution or CES function allows for other values of the elasticity of substitution among its inputs and adds the term $\theta[\ln K - \ln L]^2$ to the Cobb–Douglas function. The translog function allows for non-homotheticity and adds the quadratic and interaction terms of the inputs to the function. A number of firm-level IT studies consider the more flexible translog function in addition to the Cobb–Douglas function (Brynjolfsson and Hitt, 1995; Dewan and Min, 1997; Hempell, 2005a). The results of estimating elasticities econometrically are discussed in Section 5.2.

3.3. Other empirical approaches

We also briefly discuss further empirical approaches. A popular approach to determine the causality among variables is testing for Granger causality. Applying this approach (Granger, 1969) to the question of whether ICT causes output or productivity growth is to test how much of output/productivity in period t is explained by past values of output/productivity and then to test whether adding lagged values of ICT can improve the explanation. Usually, it is said that output/productivity is Granger-caused by ICT if the coefficients on lagged ICT are statistically significant. However, this approach is predominantly data-driven and requires sufficiently long time series, which are relatively scarce in a firm-level context. Nevertheless, Cronin et al. (1993), the most-cited study using Granger causality, suggests a feedback process between telecommunication and economic growth. Dutta (2001) also finds that the causality from telecommunications infrastructure to economic activity is stronger than the other way around.

Extending the analysis of the ICT–productivity relationship from a production theory perspective to consumer-focused approaches calls for calculating the consumer surplus originating from the implementation of new technologies. These approaches go beyond the measurement of production output as not all value is accounted for by traditional GDP measures; e.g. how to measure all the information available for free on the internet? Hausman et al. (1997) estimate the foregone consumer surplus due to the regulatory delay in introducing mobile phones to \$100 billion. Greenstein and McDevitt (2009) measure the economic value of broadband in terms of revenue and consumer surplus in the US from 1999 to 2006. They show that broadband accounts for \$28 billion of internet access revenue, while consumer surplus is at \$4.8–\$6.7 billion, which is not different in scale from other investments. Brynjolfsson and Hitt (1996b) directly compare the effect

of IT on consumer surplus, business profitability, and productivity. Their results suggest that IT increases productivity and consumer value, but shows no effect on business profits or stock prices. Brynjolfsson (1996) estimates the consumer surplus generated by computers in 1987 constituting between \$50 and \$70 billion. Brynjolfsson et al. (2003) estimates the benefit through an increased variety of books provided by online booksellers and finds that it leads to enhanced consumer welfare between \$731 million and \$1.03 billion. Bapna et al. (2008) estimate the median consumer surplus of eBay transaction at \$4 per auction.

Besides studies on the ICT effect on consumer surplus, the empirical literature also provides estimates of ICT on firm profitability. These approaches assess the role of ICT in a context of competitive advantage (Im et al., 2001; Tam, 1998). Im et al. (2001), for example, examine changes in the market value of the firm reflected by stock prices in response to IT investment announcements. They provide empirical evidence on positive stock market reactions to IT investment announcements and shed new light the effect of IT on firm performance. Particularly, smaller firms benefit from leveraging the lower price/performance ratio of new IT and reaping greater rewards from IT investments.

4. Data issues

An important issue in the empirical IT literature is the underlying data or to put it in the words of Griliches (1994): “our understanding of what is happening in our economy (...) is constrained by the extent and quality of the data”. He argues that national account data often show many inexplicable changes in productivity and puts this down largely to measurement difficulties, i.e. data problems. One reason for increasing measurement problems is the growing importance of more unmeasurable sectors, such as finance, business services or health care. Another caveat regarding the output measure in IT studies is that IT enables firms to increase output quality in the form of new products or in improvements in quality or faster response, and this is particularly hard to capture in macroeconomic studies and therefore evidence may be substantially biased downward in productivity studies (Brynjolfsson and Hitt, 2000).

The ICT measure for country- or industry-level studies use the official national account data to measure ICT capital. The most common source are BEA and BLS for the US and EU KLEMS for harmonized European data. But price deflators for IT in the early days had suffered from inappropriate measurement and usually were assumed to underestimate the quality improvements due to incomplete quality adjustments (Baily et al., 1998; Griliches, 1994). Meanwhile, US statistical agencies have invested large resources in improving their quality adjusted price indexes and Stiroh (2002b) argues that the direction of the bias of the quality adjustments is not clear. While some argue that the price decline may be overestimated, that current personal computers may have ever greater functionality, but the “basic tasks of word processing and spreadsheet analysis are not carried out all that differently” (Baily, 2002). Apart from this argument difficulties remain to ac-

Table 3

Overview of most cited studies.

ICT measure/ aggregation level	Firm	Industry	Country
Telco	–	Greenstein and Spiller (1995) (44), Correa (2006) (14)	Röller and Waverman (2001) (441), Hardy (1980) (184)
Broadband	Grimes and Ren (2009) (6)	–	Lehr et al. (2005) (73), Crandall et al. (2007) (58)
IT	Bresnahan et al. (2002) (1609), Brynjolfsson and Hitt (1996b) (1319)	Baily and Lawrence (2001) (241), Stiroh (1998) (133)	Gordon (2000) (1123), Jorgenson and Stiroh (1999) (298)
ICT	Bertschek and Kaiser (2004) (118), Hempell (2005b) (92)	Stiroh (2002b) (608), Morrison (1997) (251)	Oliner and Sichel (2000) (1417), Jorgenson (2001) (991)

Notes: First number in parenthesis indicates year of publication, the second number is the Google Citation. Index from Spring 2010. The numbers have been directly taken from the Google Scholar site and have not been adjusted for any bias, as e.g. very recent papers in 2010 will always show low numbers. Telco includes studies on wireless and wireline telecommunication. Broadband covers all the internet studies, that usually deal with broadband access. IT studies are usually based on a computer measure.

count for investments in software, especially regarding customized software. Indeed it was only in 1999 that the BEA reclassified software as an investment instead of an expense (Dedrick et al., 2003). For the integration of intangibles into growth accounts, Corrado et al. (2005) provide a framework, but the implementation and development of measuring intangible capital is still developing.

Another data issue is the scope of the ICT measure. According to Oz (2005) the IT input used in any productivity study must include: computing hardware, telecommunications hardware and software, purchased software, software development, consulting services, and personnel training. However, in many cases due to lack of data a simple PC measure is implemented (Black and Lynch, 2001; Greenan and Mairesse, 2000; Loveman, 1994), which arguably is correlated with other ICT spending. An alternative is to add to the PC measure the expenditures for IT staff (Brynjolfsson and Hitt, 1995, 1996a). Other studies differentiate between different IT products, e.g. Cappelli (2010) investigates the effect of the use of company intranets, data warehousing practices, performance support systems, and employee competency databases. Maliranta and Rouvinen (2006) split their ICT measure according to different portability characteristics and find that a simple significant computer effect can be increased with wireless connectivity and portability.

The most direct implementation of the production function is to use a IT capital stock measure. One of the most widely applied firm-level data sources, the Harte Hanks Technology Database (CITDB), formerly known as the Computer Intelligence InfoCorp (CII) database, offered an IT stock measure only until 1995 (Tambe and Hitt, 2011). Therefore firm-level studies using later data or different sources either had to approximate a capital stock measure on the basis of PC information or instead used an intensity measure such as PC per employee. Similarly, at the macro level some construct stock measures on ICT (Jorgenson and Stiroh, 2000; Oliner et al., 2007; Schreyer, 2002; van Ark et al., 2003), while other studies, especially for communication technologies, use penetration levels (Hardy, 1980; Koutroumpis, 2009; Sridhar and Sridhar, 2007).

5. Results of our survey

For an overview of existing research, Table 3 lists the most cited articles for every field of research, where a field

is determined by the aggregation level and the ICT product. As the rate of the Google Citation Index indicates, the most active fields are firm-level studies of IT on productivity and country studies on the ICT effect. The two studies in the latter field (ICT-country) both are growth accounting studies, while the most cited IT-firm and communication-country studies are productivity estimations. The most recent field is the study of broadband on productivity, while there are still no firm-level studies on voice-communication or industry studies on broadband available.

5.1. Results in growth accounting

One important indicator that most growth accounting studies report and thus can be easily compared is the contribution of ICT to labor productivity in percent. It is calculated by adding up the effect of ICT capital deepening ($\bar{v}_{K_{IT}} \Delta \ln k_{IT}$) and the TFP growth in the ICT-producing industries ($\bar{w}_{IT} \Delta \ln A_{IT}$). Both reflect the contribution of investments in ICT regardless of an additional eventual existence of spillovers. Adding up these two growth rates (in percentage points) and dividing it by labor productivity growth gives us the ICT contribution in a growth accounting framework. The results from the most influential studies on the US and the EU are reported in Table 4.⁵ The contribution of ICT was the lowest in Europe before 1995 showing only 17% of productivity growth, while the peak contribution was over 70% in the US during 1995–2000.

The basic story line from growth accounting studies for the US is that after its productivity slowdown in the pre-1995 period, when productivity growth fell behind European level, the US experienced a productivity resurgence. In particular, during the period from 1995 to 2000 the US showed high investments in IT accompanied by large productivity increases. IT capital is 1.1% points of the 4.8% output growth during 1996–1999 and 40% of TFP growth originated in IT-producing sectors. The post-1995 US resurgence was thus strongly driven by the IT-producing sector (Oliner et al., 1994). Although Jorgenson (2005) finds that the productivity growth rates were by far the highest in ICT production, he concludes that the overall contribution was limited due to the low shares of this sec-

⁵ Because of business cycles, it is common in growth accounting exercises to average over growth intervals.

Table 4

Overview of ICT contribution calculated from growth accounting exercises.

	EU in % of LP (%)	Source	USA in % of LP (%)	Source
1990–1995	17	van Ark et al. (2003)	36	Jorgenson (2001)
1995–2000	42	van Ark et al. (2003)	59–73	Jorgenson et al. (2008) Oliner et al. (2007)
2000–2005	45	van Ark and Inklaar (2005)	33–43	Jorgenson et al. (2008) Oliner et al. (2007)
2003–2007	31	European Commission (2010)		

tor in the overall economy. Nevertheless, ICT accounted for 60% of labor productivity growth in growth accounting. After 2000 the impact of ICT investment on productivity slowed, but remained strong compared to the pre-1995 period. In this period reduced TFP growth in ICT-producing industries was partly offset by the rise of TFP in ICT-using industries, especially services (Jorgenson, 2007). Therefore the US experienced continued growth due to IT, albeit smaller than in the preceding decade (Oliner et al., 2007). An alternative explanation for the TFP growth post 2000 was technological progress outside the IT sector and cyclical dynamics, which led to more cautious hiring and increased competitive pressure (Gordon, 2003; Oliner et al., 2007). Industry studies will be discussed in Section 5.3, when evaluating the evidence on the GPT hypothesis.

Europe, on the other hand, shows lower productivity growth and ICT investments post 1995, while the differential to the US has increased throughout the early 2000s (van Ark and Inklaar, 2005). While the ICT contribution in the 2000–2005 period is of the same size in Europe as in the US, the reason for the EU–US gap is in substantially lower productivity growth rate in the former. This was labeled the “Atlantic divide” by Timmer et al. (2003), who emphasize that ICT is only one explaining factor for lower productivity growth. Daveri (2002) confirms that indeed part of the EU–US gap can be attributed to new technologies. However, shares of ICT in total investment are approximately half to two thirds of US levels, despite large variations of EU member countries. Also, TFP growth is consistently lower in the EU (van Ark et al., 2003). The causes of the post-2000 productivity differential are seen in the smaller contribution of market services in the EU, particularly in the retail and finances sectors, which are typically strong ICT users (van Ark et al., 2008). The reasons are mainly seen in missing efficiency gains and less due to different ICT investment levels in these sectors (Inklaar et al., 2008).

The explanation of the gap is not mainly a lacking ICT production sector, as Colechia and Schreyer (2002) suggest by comparing OECD countries, and which is supported by the case of Australia. This country is an example for a nation that experienced strong TFP growth in ICT-using industries but negligible in the ICT-producing sector. Further, although the US ICT-producing sectors exhibited stronger growth contributions to labor productivity growth compared to the EU. The divergence in the growth contribution of EU and US ICT-using sectors is more obvious. While in the EU ICT-using sectors stayed at their pre-1995 level, the US experienced a tremendous increase in the productivity contributions of ICT-using sectors (Inklaar et al., 2003).

The successful utilization of ICT in US sectors raises the question of non-measured complementary investments like organizational capital and management quality. Bloom et al. (2010) collect micro data of US and non-US multinationals on their organization and management quality. The authors show that US firms are organized in such a way that allows them to use new technologies, like IT, more efficiently. Other approaches in this area, like Acharya and Basu (2010), incorporate some measure of intangible capital leading to a reduction of the TFP growth residual.

Despite the rigid assumptions of the growth accounting framework, the results of studies differ. The reasons for this possibly originate from different output measures, as these can rely on the product side only or can be averaged over income and product accounts. Moreover, different approaches vary by explicitly including or excluding the farm sector or consumer durable and owner-occupied housing from inputs and outputs (Baily, 2002). The outcome may also be affected by which products are included in the ICT measure, the application of different price deflators, as well as different start and end dates. Generally however, capital deepening and TFP growth show similar patterns among the studies, while interpretations partly diverge.

Another important distinguishing feature is the consideration of intangible capital. Marrano et al. (2009) treat spending on knowledge-type investments, e.g. economic competencies, innovative property, and computerized information (i.e. mainly software), as investments and find that value added rises considerably. Moreover, accounting for intangible capital increases labor productivity, while there is no significant effect on TFP. This finding of a lower TFP impact is confirmed by Oliner et al. (2007), who also control for intangible assets. However, the authors also find that ICT substantially contributed to labor productivity post-2000 even after accounting for factor utilization, adjustment costs, and intangible asset accumulation.

5.2. Results in productivity estimations

Early studies did not find evidence of ICT having a significant impact on productivity (Berndt and Morrison, 1995; Brynjolfsson, 1996; Loveman, 1994), which was explained mainly with the amount of IT capital being too small for its effect to be detected in large scale studies (Schreyer, 2001). Later on, the number of studies finding a positive significant effect on productivity increased (Bertschek and Kaiser, 2004; Black and Lynch, 2004; Bloom et al., 2010; Brynjolfsson and Hitt, 2003; O'Mahony and Vecchi, 2005; Röller and Waverman, 2001).

To compare the elasticities we conducted a rigorous search procedure to avoid biases of a simple snowball

Table 5

Overview of elasticity estimates.

Authors	Published	Elasticity	Unit	Data		Region	N/Year
				Start	End		
Black and Lynch (2001)	2001	0.05	Firm	1987	1993	AM	638
Black and Lynch (2004)	2004	0.296	Firm	1993	1996	AM	284
Bresnahan et al. (2002)	2002	0.035	Firm	1987	1994	AM	300
Brynjolfsson and Hitt (1995)	1995	0.052	Firm	1988	1992	AM	n.a.
Brynjolfsson (1996)	1996	0.044	Firm	1987	1991	AM	702
Brynjolfsson and Hitt (2003)	2003	0.058	Firm	1987	1994	AM	1324
Dewan and Min (1997)	1997	0.09	Firm	1988	1992	AM	773
Gilchrist et al. (2001)	2001	0.021	Firm	1986	1993	AM	580
Brynjolfsson and Hitt (1996b)	1996	0.048	Firm	1988	1992	AM	370
Lichtenberg (1995)	1995	0.098	Firm	1988	1991	AM	1315
Tambe and Hitt (2011)	2011	0.041	Firm	1987	2006	AM	1800
Bertschek and Kaiser (2004)	2004	0.152	Firm	2000	2000	EU	212
Bloom et al. (2010)	2010	0.015	Firm	1995	2003	EU	4809
Hempell et al. (2004)	2004	0.041	Firm	1996	1998	EU	972
Hempell (2005a)	2005	0.06	Firm	1994	1999	EU	1177
Mahr and Kretschmer (2010)	2010	0.13	Firm	2000	2008	EU	182
Hempell (2005b)	2005	0.049	Firm	1994	1999	EU	1222
Loveman (1994)	1994	-0.06	Firm	1978	1984	WW	60
Basant et al. (2006)	2006	0.115	Firm	2003	2003	AS	266
McGuckin and Stiroh (2002)	2002	0.17	Industry	1980	1996	AM	10
Stiroh (2002a)	2002	-0.071	Industry	1973	1999	AM	18
Acharya and Basu (2010)	2010	0.031	Industry	1973	2004	WW	384
O'Mahony and Vecchi (2005)	2005	0.066	Industry	1976	2000	WW	55
Venturini (2009)	2009	0.138	Country	1980	2004	EU	15
Dewan and Kraemer (2000)	2000	-0.013	Country	1985	1993	WW	36
Koutroumpis (2009)	2009	0.012	Country	2002	2007	WW	22
Madden and Savage (2000)	2000	0.162	Country	1975	1990	WW	43
Röller and Waverman (2001)	2001	0.045	Country	1970	1990	WW	21
Sridhar and Sridhar (2007)	2007	0.15	Country	1990	2001	WW	63

Notes: Region abbreviations are AM: Americas, AS: Asia, EU: Europe, WW: Worldwide. N/Year are the observations per unit per year. Bold values are all significant at the 10% level or lower.

literature search. We used the following procedure: Firstly, we conducted a keyword search combining “ICT, Computers, Internet, Broadband (Mobile) Telecommunication, Information Technology” each with “Economic Growth, Productivity” for the standard academic journals.⁶ To avoid publication bias, secondly, the same keyword search was repeated with sharing sites for working papers.⁷

In the next step, in all selected studies that had estimated an elasticity, one estimate was chosen according to the following criteria: (a) the estimation with the most general sample (i.e. total industry instead of manufacturing only), (b) the most conservative estimate regarding possible endogeneity biases (instrumental variable estimation before fixed effects before simple OLS), and (c) for

comparability reasons, we chose the most parsimonious model regarding further control variables.

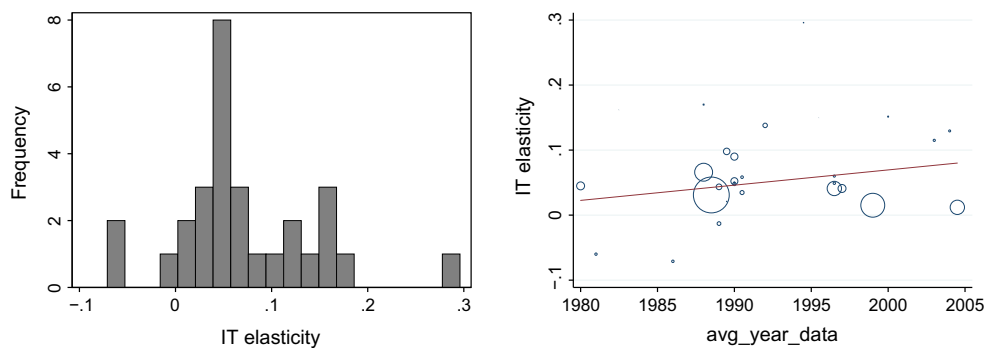
The results of the procedure are illustrated in Table 5. The majority of studies was conducted in the US, but several in Europe. According to growth accounting studies, ICT investments play a much larger role in the US, while the analyzed elasticities suggest no geographical differences in IT elasticities.

The histogram in the left panel of Fig. 1 shows a clear cluster of elasticities around the value of 0.05–0.06, but also some negative and high positive outliers. Ordering the studies by their average year of the data used for the estimation, we find a positive time trend. Other orderings by aggregation level, region or ICT measure did not yield any significant effect. Hence, the positive time trend in ICT elasticities is a clear distinguishing characteristic of our work. A positive time trend is confirmed by (Tambe and Hitt, 2012) in a study with data from 1987–2006. They estimate the marginal product of IT spending to be higher from 2000–2006 than in any previous period, suggesting that firms may have been continuing to develop new, valuable IT-enabled business process innovations.

The problem of endogeneity has already been discussed above. Interestingly, some studies show evidence of larger IT effects for application of IV estimators (Brynjolfsson, 1996; Brynjolfsson and Hitt, 2000). Tambe and Hitt (2012) finds that GMM estimators that account for endogeneity are only about 10% lower than unadjusted estimates, suggesting that the effects of endogeneity on IT productiv-

⁶ The list of the considered journals is: Academy of Management Journal, American Economic Review, Brookings Paper on Economic Activity, Econometrica, Economic Inquiry, Economic Journal, Economic Policy, Economica, Economics of Innovation and New Technology, Information & Management, Industrial and Corporate Change, Information Economics and Policy, Information Systems Research, Journal of Applied Econometrics, Journal of Business, Journal of Econometrics, Journal of Economic Growth, Journal of Economic Literature, Journal of Economic Perspectives, Journal of Industrial Economics, Labour Economics, Journal of Management Information Systems, Journal of Political Economy, Journal of Productivity Analysis, Management Science, MIS Quarterly, Organization Science, Quarterly Journal of Economics, RAND Journal of Economics, Review of Economic Studies, Review of Economics and Statistics, Review of Income and Wealth, Strategic Management Journal, Telecommunications Policy.

⁷ The following sites were included: NBER, SSRN and Groningen Growth and Development Center.



Notes: The circle sizes depend on the precision of each estimate (the inverse of its within-study variance)

Fig. 1. Frequency and variation in time of elasticities.

ity estimates are rather small. Bloom et al. (2010) find no qualitatively different results on their IT elasticity when conducting robustness checks with System-GMM and Oley Pakes estimation to control for possible endogeneity. Hempell (2005a) finds a clear exaggeration of the ICT coefficient if the endogeneity issue is neglected.

More clear is the evidence regarding the inclusion of a fixed effects. Including fixed effects yields a more conservative estimate as it discards firm-specific productivity effects that are not triggered by ICT investment, but it also discards any benefits of IT that are persistent at the firm level over time. Brynjolfsson and Hitt (1995), in fixed effects estimations, obtain a 50% lower ICT effect. Other studies have confirmed the strong downward effect when including fixed effects (McGuckin and Stiroh, 2002; Stiroh, 1998).

Other studies stress the importance of organizational complements for IT productivity. Bresnahan et al. (2002) argue that firm-level changes realize through a “three way cluster of complementarity”. IT use is more likely to be effective with a higher quality of service output mix, decentralized decision making and skilled workers. Black and Lynch (2001) also show the importance of workplace organization for IT productivity. An interesting differentiation to their finding of a positive and significant elasticity comes from their study as it shows that the greater the proportion of non-managerial workers who use computers, the higher is the plant productivity. In contrast, the proportion of managers who use computers is not significant. This points to the fact that the usage of IT at mid- to low-level workers has a decisive impact on firm performance. Crespi et al. (2007) show that including organizational capital reduces IT returns which implies a complementarity. In contrast to these findings Bertschek and Kaiser (2004) find that the individual output elasticities of ICT investment do not significantly differ between firms with or without workplace reorganization.

Another discussion around ICT productivity deals with the existence of a threshold value, which has been especially prevalent in the communication technology literature. Röller and Waverman (2001) states that the significant relationship between telecommunication and growth only holds when services reach a certain threshold near universal levels. Similarly Dewan and Kraemer (2000) finds no significant relationship for the subsample of

developing countries, nor does Shiu and Lam (2008) for China, except in affluent regions. Gruber and Koutroumpis (2011) state that the contribution of mobile telecommunication is significantly smaller for low penetration countries than for high penetration countries. This suggests that non constant returns to scale exist, at least for communication technologies.

5.3. Evidence on the GPT hypothesis

This section discusses whether the GPT hypothesis passes empirical testing. Jorgenson (2001), as one of the most important academics in the field of growth accounting models is sceptical about the existence of spillovers and argues that the rewards are large because of the “swift pace of technical change in the production of computers and the rapid deployment of IT equipment through substitution” and repeats this in his subsequent review (Jorgenson et al., 2008). Of course, the absence of spillovers is a basic assumption for the growth accounting model. Similarly, Oz (2005) argues that IT gains may have been accrued in its early days (when IT, especially software, was hard to measure), but today a firm without PC will simply not survive. Gordon (2000) states that based on his growth accounting exercises the productivity resurgence was driven by ICT production and can be seen only in the sector of manufacturing durable goods, for “the remaining 88% of the economy, the New Economy’s effects on productivity growth are surprisingly absent, and capital deepening has been remarkably unproductive”. Therefore no structural acceleration throughout the economy in productivity during 1995–1999 took place, and thus the pervasiveness of the technology, which is constituent for a GPT, is absent. In a later work Gordon (2003) adds that the effects from investment in IT has been largely exaggerated.

5.3.1. Innovation spawning and pervasiveness

Section 2 mentioned that a GPT technology is characterized by inducing further innovation and being adopted in wide scope of sectors of an economy. On the innovation spanning character of ICT, Hempell (2005a) found that innovative experience plays an important role regarding the successful use of ICT while this innovative experience is not important for conventional investments. This

Table 6
Studies on GPT hypothesis.

Approach	GPT Hypothesis confirmed	Refuted
Analyzing industry data	Stiroh (2002b), Baily and Lawrence (2001), and Bosworth and Triplett (2003)	
TFP regression with lagged ICT variable as explanatory	Brynjolfsson and Hitt (2003), Basu et al. (2003), Basu and Fernald (2007), and Greenan and Mairesse (2000)	Stiroh (2002a), van Ark and Inklaar (2005), Wolff (2002), and Inklaar et al. (2008)
Comparing with other GPTs of the past	Crafts (2002) and Jovanovic and Rousseau (2005)	Gordon (2000)
Excess return	Brynjolfsson and Hitt (1995), Lichtenberg (1995), Gilchrist et al. (2001), and O'Mahony and Vecchi (2005)	Stiroh (2005)

suggests that ICT is a “special” capital good, as unlike other goods the success is strongly linked to innovation and re-engineering of processes. [Becchetti et al. \(2003\)](#) find with Italian micro data that telecommunication investment positively affects the creation of new products and processes. The impact of broadband on firms' labor productivity is highly heterogeneous, but the impact on innovation is positive significant and robust ([Bertschek et al., 2011](#)).

Despite the large variation of ICT intensity in household adoption, often called the “digital divide” (for explanations of different demand structures see [Savage and Waldman, 2009](#)), [Jovanovic and Rousseau \(2005\)](#) show that households adopted electricity about as rapidly as they are adopting the PC and that these disparities are not unique for digital technologies. At the business level ICT investments made up constantly more than 20% of overall investments in the 1990s in the US and above 15% in the UK ([Colecchia and Schreyer, 2002](#)). In fact, [Triplett and Bosworth \(2006\)](#) argue that comparing old IT studies to more recent ones, the only real change is that ICT capital is much larger than it once was and not surprisingly contributes more to recent growth than it did in earlier periods. Even though ICT investment levels vary greatly among sectors, the ICT phenomenon is not confined to a narrow sector and finds broad range of applications, e.g. transport ([Hubbard, 2009](#)), health services ([Athey and Stern, 2002](#)) or banking ([Autor et al., 2002](#)). Hence, investments and application of ICT have been widespread, although full diffusion in every firm or household is still far from complete.

These studies suggest certain characteristics of ICT that are associated with GPT. However, a number of studies tackle this question more specifically. [Table 6](#) presents an overview of the main approaches to empirically assess the GPT hypothesis and lists the studies that applied the approaches according to their main outcome. In the following paragraphs we will discuss the outcomes in more detail.

5.3.2. Studies analyzing industry data

In an industry study [Stiroh \(2002b\)](#) shows that industries above the median in their IT intensity had about 2% points larger acceleration in labor productivity growth post-1995 than other industries. He applies a difference-in-difference approach to show that the IT phenomena was much more widespread and that the productivity surge was not solely limited to the ICT-production sector, and that it cannot be explained entirely by cyclical factors. However, he makes no judgment on the spillover property of ICT. Instead he states that the “strong and robust

correlation between IT intensity and the subsequent productivity acceleration, however, implies that there may be a deeper relationship between IT investment and productivity growth”. Similarly [Baily and Lawrence \(2001\)](#) find substantial acceleration in labor productivity outside the computer sector with a labor productivity measure (gross domestic income per employee) that incorporates both capital deepening and TFP growth and they interpret their findings in support of the GPT hypothesis. [Bosworth and Triplett \(2003\)](#) calculate labor productivity and TFP by industry and find that the accelerating productivity in the service sector plays a crucial part in the productivity resurgence post 1995. They state that particularly service sectors use ICT intensively, providing evidence of ICT usage and TFP growth. In a later study [Bosworth and Triplett \(2007\)](#) confirm that non-ICT-producing sectors saw a sizable acceleration in TFP, especially in the service industries during the 2000s, whereas post-2000 TFP growth in ICT-producing sectors in fact declined compared to its golden era in the mid-1990s. Other, more descriptive studies therefore indicate that ICT triggers innovation in the ICT-using sector. All of these studies have been conducted with US data.

5.3.3. TFP regression studies

Regarding spillovers, in the case of communication technologies network effects are intuitive. Most studies simply state this fact as a basic premise of telecommunication ([Hardy, 1980](#); [Leff, 1984](#)) or for broadband ([Majumdar et al., 2009](#)). For ICT in general and particularly for computerization, the existence of spillovers is questioned. Vertical spillovers would imply that TFP growth must be evident also in ICT-using industries. Therefore one important missing link would be to find hard empirical evidence of spillovers, which could be found most directly if ICT use or investments have a positive significant effect on TFP.

As stated in [Section 3.1](#), TFP is an indicator of innovation in an industry. The corresponding testing equation regresses ICT investment on TFP. But finding a positive effect has to be interpreted with great care, as it may be explained by true spillovers, i.e. due to fundamental improvements in the production process as a consequence of investing in ICT, but it may also be due to mismeasurement and consequently a spurious sign of spillovers. Firstly, as already discussed in [Section 3.1](#) TFP is highly prone to mismeasurement. Secondly, organizational or other forms of intangible capital are not measured and therefore not all inputs are observed correctly.

As investment in organizational capital is correlated to ICT, without accounting for complementary investments in the regression equation, the ICT effect will be biased. Of course, the huge complementary investments in form of intangible capital are also an indication for a GPT, but for TFP regressions this clearly leads to an overstatement. Further, as it takes time for externalities to materialize TFP should rise in ICT-using industries with a lag. In fact, contemporaneous investments in ICT may be associated with lower TFP as resources are diverted to reorganization and learning (Basu et al., 2003). But the exact timing is unknown and no theory is available to determine which exact positive lag would prove the existence of true spillovers. Finding the appropriate lag therefore is an empirical question. And finally, the problem of endogeneity is exacerbated as the dependent variable was constructed through the explanatory variable, ICT capital (Draca et al., 2008).

TFP regressions are usually operationalized at the industry level. Stiroh (2002a) formally tests for spillovers with an ICT dummy that divides industries by the median of ICT capital share, and finds no significant link, implying the absence of spillovers or that the “benefits of ICT investment are consistent with the costs”. Wolff (2002) adds to this evidence, as he also finds no econometric evidence that computer investment is positively linked to TFP growth. But in further regressions he does find that IT is positively associated with change, such as occupational restructuring and changes in the composition of inputs and capital coefficients. van Ark and Inklaar (2005) find no positive significant effect in their TFP regressions, but do find a clear u-shaped pattern, when estimating the effect for 5-year periods separately for the years 1979–2003. This pattern – though not significant – can be reconciled well with the GPT theory, specifically with the reasoning that economic gains accrue only gradually. Therefore the u-shaped curve shows normal returns for the first years, followed by negative effects in a period of substantial co-investment, which only when the combination of the investments comes into effect can ultimately lead to a positive effect of ICT on TFP. In a later similar study, again, no spillovers were found (Inklaar et al., 2008).

Siegel and Griliches (1992) find a positive and statistically significant relationship between TFP growth and an industry's rate of investment in computers, though they point to some data reliability problems. Basu et al. (2003) show in their study that lagged ICT capital growth is significantly positively correlated with TFP growth in the US, but not in the UK, which they interpret as indication of the GPT nature of ICT. An updated version with US data confirmed the results and shows that ICT capital growth with long lags is positively associated with industry TFP acceleration (Basu and Fernald, 2007). However, they have been criticized for their “ad hoc” choice of periods and lags (van Ark and Inklaar, 2005). Nevertheless, their results are corroborated by Brynjolfsson et al. (2003), who show with TFP regressions at the firm level that the effects are stronger when IT is measured over longer time periods. Productivity and output contributions associated with computerization are up to five times greater over long periods. They interpret this as a sign of complementary investments in organizational capital, which are reflected in the IT coefficients.

Even if it mainly reflects unmeasured investments, it shows that IT is related to a broader set of assets and “that the long-run contribution of computerization to growth is potentially much larger than would be expected from the quantity of direct investment in computer capital”. Greenan and Mairesse (2000) also find a positive and significant effect of ICT on TFP for French firm-level data, though these effects are markedly smaller than the ICT effect on labor productivity. Summarizing, the TFP regressions, which would be the most clearest and direct piece of evidence for the GPT theory, show somewhat ambiguous results which may be related to their methodological weaknesses.

Recent studies have extended simple TFP regressions to better identify whether the ICT coefficient displays real spillover effects, again with varying outcomes. Acharya and Basu (2010) argue that if the positive ICT term on an industry level suggests true spillovers, this effect should prevail for cross-border ICT investments of the same industry. In this study no evidence of positive spillovers to ICT investment are found, neither within countries nor across borders. Castellacci (2010) follows a very different approach and identifies sectors according to their technological characteristics, and categorizes them into providers or recipients of advanced knowledge showing that sectoral groups that are close to ICT have improved productivity performance.

5.3.4. Comparing ICT with GPT technologies of the past

Another approach is to compare ICT with other widely accepted general purpose technologies. This is usually done by compiling a number of descriptives and anecdotal evidence. Crafts (2002) repeats the growth accounting exercise for electricity's contributions to US growth for the year 1899–1929 and for steam engine in the UK for the year 1760–1860. He concludes that the growth contribution of ICT in the past 25 years has exceeded that of steam, and at least it matched that of electricity over a comparable period. Jovanovic and Rousseau (2005) document key differences between the diffusion paths of electricity and IT and analyze, based on a number of facts, their GPT characteristics. They conclude that the two GPT eras are strikingly similar in a number of respects and that IT is the more “revolutionary” of the two GPTs, even though its full impact is yet to be seen.

5.3.5. Excess returns of ICT

A final approach to assess the GPT nature of ICT in our study is the investigation of excess returns to ICT. In the simplest case firms invest in ICT until the marginal product of ICT equals its rental price. To determine whether the ICT coefficient is excessive or greater than those earned by other assets, the estimated marginal product has to be compared with its rental price or the elasticity with its revenue share. The revenue share rises with the amount of ICT capital invested and with the rental price. Hence, high elasticities and marginal products may just reflect the high depreciation rates associated with ICT capital. Other reasons for finding excess returns may be due to omitted variables measurement error, simultaneity bias or large adjustment costs that have to be covered when investing in ICT (Stiroh, 2005).

Several microeconomic studies find excess private returns. Gilchrist et al. (2001) finds that the IT effect is above the cost share, but only in the durable goods manufacturing sector, which actually would confirm the view by Gordon (2000) that the ICT effect is limited to only a few industries. O'Mahony and Vecchi (2005) use a pooled mean estimator to receive long-run ICT effects, which they find to be above its cost share. To argue that the excess returns are an indicator for spillovers rather than private returns (excess private returns do not necessarily indicate a positive aggregate effect on the economy as firms that invest in ICT may merely displace those that do not), Brynjolfsson et al. (2003) draw upon anecdotal evidence. However, there is no empirical evidence to date for “horizontal” spillovers to other firms (Draca et al., 2008), which would indicate that the ICT investments of other firms (in the region or in the industry) have a positive significant effect on the firm's productivity. Stiroh (2005) calculates the average revenue share with his dataset and comparing it with the estimate of 0.05, he classifies the result as a normal return.

5.3.6. Weighing the evidence on GPT

Unlike the regressions on investment in ICT and labor productivity, which overwhelmingly show a positive and significant effect, the empirical evidence for ICT as GPT is more contradictory. One problem to obtaining the last convincing piece of evidence are the caveats regarding the results of the TFP regressions. Before the endogeneity issue or the question of the lag structure is solved, it is unlikely that a unanimous verdict will emerge. However, ICT very convincingly displays many GPT characteristics. So even if the ultimate piece of evidence is missing, it is still a strong circumstantial case or, to put in the words of Basu et al. (2003): “the suspect's fingerprints are all over the crime scene”.

6. Conclusions

Our survey gives an overview of the vast empirical literature on ICT and productivity and highlights the main numerical results and the methodological differences. Overall, the empirical studies demonstrate that ICT not only plays an important role in everyday lives, but in the productivity statistics as well. The evidence further indicates that the productivity effect is not only significant and positive, but also increasing over time. Of course however, having a significant effect does not mean that low performers can increase productivity by simply increasing ICT investment. ICT has to be embedded in complementary organizational investments, skills and industry structures.

We find strong indication but no final evidence that ICT is a General Purpose Technology. Interestingly, the existing positive evidence was mainly found for US data, while evidence for European countries is more ambiguous. Therefore, a better understanding of how spillovers work with ICT might help to bridge the gap, because many questions concerning possible externalities remain unanswered: do management ideas and knowledge on ICT diffuse among firms? How can this knowledge best be transferred? Which

time lag is needed for the spillovers to materialize? Further, while the GPT property of ICT is widely discussed and empirically tested, GPT properties has not been tested explicitly for broadband, although this seems to be the main target of policy agendas. Finally, the endogeneity issue still remains unsolved. Thanks to the existence of computerization the possibilities to empirically investigate these questions have improved greatly, which should help us to continue to empirically observe how exactly ICT is changing the economy.

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